Code No: 157EH **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
BASIC MECHANICAL ENGINEERING
(Common to EEE, CSE, IT)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define black body.	[2]
b)	Define point and path functions.	[3]
c)	What is degree of saturation?	[2]
d)	Explain the purpose of carburetor in SI engine.	[3]
e)	Define interference.	[2]
f)	What is a differential gear? Where is, it used?	[3]
g)	Define mechanical advantage.	[2]
h)	Define the terms prime circle, pitch circle, base circle of a cam.	[3]
i)	Define resonance.	[2]
j)	List the advantages of multistage compressors.	[3]

PART – B

(50 Marks)

2.a) How does transient heat transfer differ from steady heat transfer? How does one-dimensional heat transfer differ from two-dimensional heat transfer?
b) No engine can be made to work on Carnot cycle" justify the statement. [5+5]

OR

3.a) Define thermodynamic equilibrium. Explain different laws of thermodynamic.
b) Discuss the different modes of heat transfer. [5+5]

4.a) What are the main components of automobile which contribute for efficient functioning? Explain.
b) What is the difference between vapour compression cycle and absorption cycle? [5+5]

OR

5.a) Why compression ratio of petrol engine is low compared to diesel engines?
b) Explain the vapour compression cycle with a neat sketch. [5+5]

6.a) Derive an expression for minimum number of teeth required on a pinion to avoid interference when it gears with a rack.
b) Why are hydraulic systems preferred for heavy work than the pneumatic systems? [5+5]

OR

7.a) What are the laws of gearing? Explain.
b) What are the basic concepts of hydraulic power? Explain. [5+5]

8.a) Sketch and define transmission angle of a four-bar mechanism.
b) What is the differences between an element and a kinematic link of a mechanism? How do you classify links of a mechanism? [5+5]

OR

9.a) Define the following:
i) Kinematic chain
ii) Mechanism
iii) Structure.
b) Derive the expression for velocity and acceleration during out-stroke and return stroke of the follower. [5+5]

10.a) Explain the working principle of centrifugal compressor.
b) Differentiate between free vibrations and forced vibrations. [5+5]

OR

11.a) With a neat sketch, explain the parts of Reciprocating compressor.
b) Define the following terms:
i) Periodic motion
ii) Fundamental mode of vibration
iii) Degree of freedom
iv) Simple harmonic motion [5+5]

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